CHAPTER 2

ELECTROSTATIC POTENTIAL AND CAPACITANCE

CHAPTER ESSENCE AT A GLANCE:
Electrostatic potential energy, equipotential surfaces, electric field gradients, and energy storage in capacitors with dielectrics.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Electrostatic Potential & Conservative Work:** Electric potential represents the work done per unit positive charge in bringing it from infinity against electrostatic forces, completely independent of the path chosen.

• **Equipotential Surfaces & Field Orthogonality:** Surfaces of constant electric potential where electric field vectors are everywhere perpendicular to the surface, meaning zero work is done moving a charge along an equipotential surface.

• **Capacitance & Dielectric Polarization:** Capacitance quantifies charge stored per unit potential difference; inserting a dielectric medium reduces internal field and increases capacitance by the dielectric constant.

• **Electrostatic Energy Storage:** Electrical energy in a charged capacitor resides directly within the electrostatic field established between the plates.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$V = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

Electrostatic potential due to a point charge

$$U = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r_{12}}$$

Electrostatic potential energy of two point charges

$$C = \frac{\epsilon_0 A}{d}$$

Capacitance of parallel plate capacitor in air/vacuum

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV$$

Energy stored in charged capacitor

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Battery Disconnected: Charge Q remains CONSTANT; V and U decrease by factor K. Battery Connected: Potential V remains CONSTANT; Q, C, and U increase by factor K.

• **Alert:** Work done in moving a test charge across an equipotential surface is ZERO because displacement is perpendicular to electric force.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Electric potential due to a point charge [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Principle & Definition]** Electric potential V at point P is the work done in bringing a unit positive test charge from infinity to point P against electrostatic forces without acceleration: $V = -\int_{\infty}^{r} E \cdot dr$. [1 Mark]

• **[Step 1: Work Done Derivation]** For point charge +q at origin, electric field at distance x is $E = (\text{one-fourth } (0.25)\pi\varepsilon_0)\ (q \text{ divided by } x^2)$ unit vector r. Work done moving test charge $+q_0$ from x to x + dx: $dW = F \text{ ext} \cdot dx = -(\text{one-fourth } (0.25)\pi\varepsilon_0)\ (q\ q_0 \text{ divided by } x^2)\ dx$. $W = -(q\ q_0 \text{ divided by } 4\pi\varepsilon_0) \int_{\infty}^{r} (1 \text{ divided by } x^2)\ dx = -(q\ q_0 \text{ divided by } 4\pi\varepsilon_0)\ [-1 \text{ divided by } x]_{\infty}^{r} = (q\ q_0 \text{ divided by } 4\pi\varepsilon_0)\ [1 \text{ divided by } r - 0] = (\text{one-fourth } (0.25)\pi\varepsilon_0)\ (q\ q_0 \text{ divided by } r)$. [1.5 Marks]

• **[Step 2: Potential Formula]** $V = W \text{ divided by } q_0 = (\text{one-fourth } (0.25)\pi\varepsilon_0) \cdot (q \text{ divided by } r)$. [0.5 Mark]

• **Key Exam Result:** $V = (\text{one-fourth } (0.25)\pi\varepsilon_0) \cdot (q \text{ divided by } r)$; Scalar quantity, Unit: Volt ($V = J\ C^{-1}$)

2. Electric potential on axial and equatorial points of dipole [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup]** Dipole of charges -q at A(-a) and +q at B(+a). Dipole moment p = q(2a). [1 Mark]

• **[Case 1: Axial Line (distance r from center O)]** $V \text{ axial} = V_+ + V_- = (q \text{ divided by } 4\pi\varepsilon_0)\ [1 \text{ divided by } (r - a) - 1 \text{ divided by } (r + a)] = (q \text{ divided by } 4\pi\varepsilon_0)\ [2a \text{ divided by } (r^2 - a^2)] = (\text{one-fourth } (0.25)\pi\varepsilon_0)\ [p \text{ divided by } (r^2 - a^2)]$. For short dipole (r >> a): $V \text{ axial} = (\text{one-fourth } (0.25)\pi\varepsilon_0) \cdot (p \text{ divided by } r^2)$. [1 Mark]

• **[Case 2: Equatorial Line]** Field point P is equidistant from both charges: $AP = BP = \text{sqrt}(r^2 + a^2)$. $V \text{ eq} = V_+ + V_- = (\text{one-fourth } (0.25)\pi\varepsilon_0)\ [q \text{ divided by } \text{sqrt}(r^2+a^2) - q \text{ divided by } \text{sqrt}(r^2+a^2)] = 0$. Electric potential is IDENTICALLY ZERO at all points on the equatorial plane of a dipole! [1 Mark]

• **Key Exam Result:** $V \text{ axial} = (\text{one-fourth } (0.25)\pi\varepsilon_0) \cdot (p \text{ divided by } r^2)$ | V equatorial = 0 at all points

3. Equipotential surfaces: definition and properties [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Definition]** An equipotential surface is a surface having the same electric potential at every point on it (V = constant). [1 Mark]

• **[Step 2: Four Key Properties]** 1. Work done in moving a charge between any two points on an equipotential surface is zero: $W = q\ \Delta V = q(0) = 0$. 2. Electric field E is ALWAYS normal (perpendicular) to the equipotential surface at every point (if E had a parallel component, work would be non-zero). 3. Equipotential surfaces never intersect each other (intersection would mean two different potentials at the same point). 4. Spacing of equipotential surfaces indicates field strength: $E = -dV \text{ divided by } dr$ implies $dr = -dV \text{ divided by } E$; closer spacing corresponds to stronger electric field. [2 Marks]

• **Key Exam Result:** Equipotential: V=const; W=0; E perpendicular to surface; spacing dr proportional to 1 divided by E

4. Relation between electric field and potential gradient E = -dV divided by dr [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Work Done Analysis]** Consider two closely spaced equipotential surfaces A and B with potentials V and V + dV separated by distance dr. Work done in moving a unit positive charge from surface B to surface A against electric field E: $dW = F \text{ ext} \cdot dr = -E\ dr$. [1 Mark]

• **[Step 2: Equating to Potential Difference]** By definition of potential difference: $dW = V_A - V_B = V - (V + dV) = -dV$. Equating work expressions: $-E\ dr = -dV$ implies $E = -dV \text{ divided by } dr$. Physical significance: 1. Negative sign indicates electric field points in the direction of steepest decrease of potential. 2. Magnitude of E equals rate of change of potential with distance (potential gradient). [1 Mark]

• **Key Exam Result:** $E = -dV \text{ divided by } dr$ (Electric field is negative gradient of electric potential)

5. Potential energy of a system of two point charges [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Work Done Derivation]** Bring charge q_1 from infinity to position r1: $W_1 = 0$ (no initial electric field). Electric potential at position r2 due to q_1: $V_1(r2) = (\text{one-fourth } (0.25)\pi\varepsilon_0) \cdot (q_1 \text{ divided by } r_{12})$. Work done bringing charge q_2 from infinity to position r2 against field of q_1: $W_2 = q_2 \cdot V_1(r2) = (\text{one-fourth } (0.25)\pi\varepsilon_0) \cdot (q_1\, q_2 \text{ divided by } r_{12})$. [2 Marks]

• **[Step 2: Potential Energy Formula]** Total work stored as electrostatic potential energy: $U = W_1 + W_2 = (\text{one-fourth } (0.25)\pi\varepsilon_0) \cdot (q_1\, q_2 \text{ divided by } r_{12})$. If charges have same sign ($q_1\, q_2 > 0$), U > 0 (repulsive system). If opposite sign, U < 0 (bound or attractive system). [1 Mark]

• **Key Exam Result:** $U = (\text{one-fourth } (0.25)\pi\varepsilon_0) \cdot (q_1\, q_2 \text{ divided by } r_{12})$ (Unit: Joules J)

6. Numerical on series/parallel capacitor network [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Given Circuit & Reduction]** Network with $C_1 = 2\ \mu F$, $C_2 = 3\ \mu F$ in parallel, connected in series with $C_3 = 5\ \mu F$ across 100 V battery. [1 Mark]

• **[Step 2: Calculations]** 1. Parallel combination: $C\ p = C_1 + C_2 = 2 + 3 = 5\ \mu F$. 2. Equivalent capacitance: $C\ eq = (C\ p \cdot C_3) \text{ divided by } (C\ p + C_3) = (5 \times 5) \text{ divided by } (5 + 5) = 2.5\ \mu F$. 3. Total charge from battery: $Q\ total = C\ eq \cdot V = 2.5\ \mu F \times 100\ V = 250\ \mu C$. 4. Voltage across C_3: $V_3 = Q\ total \text{ divided by } C_3 = 250\ \mu C \text{ divided by } 5\ \mu F = 50\ V$. 5. Voltage across parallel block: $V\ p = 100 - 50 = 50\ V$. Charge on $C_1 = C_1\ V\ p = 2\ \mu F \times 50\ V = 100\ \mu C$; Charge on $C_2 = 3\ \mu F \times 50\ V = 150\ \mu C$. [2 Marks]

• **Key Exam Result:** $C\ eq = 2.5\ \mu F$; $Q\ total = 250\ \mu C$; $V_1 = V_2 = 50\ V$, $V_3 = 50\ V$

7. Numerical on dielectric slab insertion [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula for Dielectric Slab of Thickness t < d]** Capacitance with slab of thickness t and dielectric constant K: $C = \varepsilon_0\ A \text{ divided by } [\ (d - t) + t \text{ divided by } K\ ]$. [1.5 Marks]

• **[Step 2: Numerical Evaluation]** Given plate area $A = 100\ cm^2 = 10^{-2}\ m^2$, plate separation $d = 4\ mm = 4 \times 10^{-3}\ m$, slab thickness $t = 2\ mm = 2 \times 10^{-3}\ m$, K = 4. Effective separation $d' = (d - t) + t \text{ divided by } K = (4 - 2)\ mm + (2\ mm \text{ divided by } 4) = 2\ mm + 0.5\ mm = 2.5\ mm = 2.5 \times 10^{-3}\ m$. $C = (8.854 \times 10^{-12} \times 10^{-2}) \text{ divided by } (2.5 \times 10^{-3}) = 3.54 \times 10^{-11}\ F = 35.4\ pF$. (Without slab, $C_0 = 22.1\ pF$). [1.5 Marks]

• **Key Exam Result:** $C = \varepsilon_0\ A \text{ divided by } (d - t + t \text{ divided by } K) = 35.4\ pF$